INCH-POUND

MIL-F-51539A(EA)
28 September 1990
SUPERSEDING
MIL-F-51539(EA)
29 March 1988

MILITARY SPECIFICATION

FILTER UNIT, GAS-PARTICULATE: 400 400 CFM , 208 V, 400 Hz, M159

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of gas particulate filter unit capable of delivering filtered air at a rate of 400 cubic feet per minute (cfm). (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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SPECIFICATIONS

MILITARY

MIL-F-51527-- Filter Set, Gas and Pparticulate, 200 CFM (Modular Collective Protcetion Equipment (MCPE)/Simplified Collective Protection Equipment (SCPE) Applications)

STANDARDS

MILITARY

MIL-STD-781 - Reliability Testing for Engineering Development, Qualification, and Production

HANDBOOKS

MILITARY

MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 191 11-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-6699 - Filter Unit, Gas-Particulate, 400 CFM, 208 V, 400 Hz, M59

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-6699 - Filter Unit, Gas-Particulate: 400 CFM, 208 V, 400 Hz, M59

(Copies are available from Commander, U.S. Army information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61229 -7300.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing, in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-19-6699 and subsidiary drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the filter unit shall conform to the specifications and drawings listed on Drawing 5-19-6699 and subsidiary drawings.

3.2 Manufacture and assembly. The filter unit shall be manufactured and assembled in accordance with Drawing 5-19-6699.

3.3 Performance.

3.3.1 Cycling bum-in. The filter unit shall be cycled for a period of 3 hours during which time the airflow valve shall drive alternately to the fully open and then to the fully closed position (see 6.3) at a rate of 3 times per minute in response to an air pressure differential of 25 inches water gage (iwg) when tested in accordance with 4.4.6.2.

3.3.2 Operational characteristics. Prior to and after completion of the cycling bum-in, the filter unit shall meet the following operational characteristics:

3.3.2.1 Airflow valve closing. The airflow valve shall start to drive closed between 1.2 and 1.7 iwg when tested in accordance with 4.4.6.3(b).

3.3.2.2 Airflow valve opening. The airflow valve shall start to drive open between 1.2 and 1.7 iwg when tested in accordance with 4.4.6.3(c).

3.3.2.3 Warning horn and MASK indicator. The warning horn shall sound and the MASK indicator shall flash between 0.85 and 1.10 iwg when tested in accordance with 4.4.6.3(c) and 4.4.6.3(d).

3.3.2.4 Fan assembly rotational speed. The fan assembly impeller rotational speed shall be a minimum of 11,000 rpm when tested in accordance with 4.4.6.3(e).

3.4 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.3.

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3.5 Reliability. Concurrent with the start of regular production, a production sample of four filter units shall be tested in accordance with Task 302 of MIL-STD-781 and test plan IV-D of MIL-HDBK-781. The lower test mean-time-between-failure (0_1) shall be 467 hours and the upper test mean-time-between-failure (0_0) shall be 935 hours. The production sample shall be tested continuously in the configuration specified in 4.4.6.1 and 4.4.6.2 without temperature, vibration or input voltage cycling. An accountable failure shall be defined as any deviation from the requirements specified in 3.3.2.

3.6 Workmanship. The filter units shall be free from cracked, bent or dented metal sections and foreign matter (dirt, oil or viscous materials).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.5.) When

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required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in this specification or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.3.2 Sample. The first article sample shall consist of five filter units manufactured and packaged (when applicable) using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.3 Inspection procedures. The first article sample of filter units shall be examined and tested for compliance with all the requirements of the applicable drawings, and sections 3 and 5 of this specification except for requirement 3.5.

4.3.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and at essentially the same time.

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4.4.2 Sampling.

4.4.2.1 For examination. Sampling for examination shall be conducted in accordance with the classification of characteristics in 4.4.5 and table I.

4.4.2.2 For test. Each filter unit shell be tested in accordance with 4.4.6. in addition, a production sample of four filter units shall be randomly selected from the first month's production. This sample shall be tested separately from any other units in the lot.

4.4.3 inspection procedure.

4.4.3.1 For examination. Sample filter units shall be examined in accordance with the classification of characteristics in 4.4.5 and table I.

4.4.3.2 For test. Each unit shall be tested for cycling burn-in and operational characteristics using the procedures set forth in 4.4.6. Any unit which fails to meet the requirements for the cycling burn-in (3. 3.1) or any of the operational characteristics (3.3.2) shall be rejected. The production sample of four filter units shall be tested in accordance with 4.4.6 for conformance to 3.3.2. Then the filter units shall be tested continuously in the configuration specified in 4.4.6.1 and 4.4.6.2 for conformance to 3-5. The sample shall meet the requirements of 3.5 and the criteria of test plan IV-D of MIL-HDBK-781 to be acceptable.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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TABLE I. Sampling

Lot size	inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.

Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
1.4.5(a)	Filter unit, gas-particulate: 400 cfm, 208 V, 400 Hz, M59		5-19-6699
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Component evident, correct and correctly assembled	Table I, level VII	3.1	CE
102	Cable undamaged	Table I, level VII	3.1	CE and Vi
103	Color dots evident and correct	Table I, level VII	3.1	VI
104	Gaskets undamaged	Table I, level VII	3.1	CE and VI
105	Workmanship	Table I, level VII	3.6	VI

NOTES:

CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER P5-19-6699
4.4.5(b)	Packaging		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Item completely clean and dry prior to unit packing	Table I, level IX	5.2	VI
102	Packaging assembly evident and correct	Table I, level IX	5.2	CE and VI
103	Marking evident, correct, and legible	Table I, level IX	5.2	CE and VI

NOTES:

CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Tests. Cycling burn-in and operational characteristics tests shall be conducted as follows:

4.4.6.1 Preparation for tests. To conduct the following test procedures, a framework of wood (or other suitable material) shall be fabricated to simulate placement of the filters within the filter unit housing. With the framework so emplaced, care should be taken to ensure that the bolts on the outer access cover and the spring loaded retaining bolt on the yoke are not overtorqued such that component damage results.

4.4.6.2 Cycling burn-in. The filter unit need not be assembled. Test the filter unit, less gas-particulate filters, by electrically interconnecting the modules for the specified time as follows:

(a) Connect the filter unit to a source of 208 V, 400 Hz, 3-phase wye power, and turn the power switch to the ON position.
(b) Connect the hose fitting on the compartment control module to a solenoid actuated vacuum source of 25 ± 2 iwg. The solenoid shall be such that it will alternatively apply the 25 iwg vacuum source and ambient pressure to the fitting.
(c) Connect the coil of the solenoid to a timer so that the solenoid will actuate at a rate of 3 times per minute (1 time per minute for the production reliability test, 3 times per minute for the burn-in). Operate the filter unit in this configuration for the specified time causing the airflow valve to alternately drive fully open and fully closed in response to the applied pressure signals.
(d) Operate the filter unit for 30 minutes. Then turn the power switch to the OFF position for 1 minute.
(e) Continue to alternate the power switch ON 30 minutes, OFF 1 minute for the specified bum-in period.

4.4.6.3 Operational characteristics. Prior to and after the cycling burn-in test, the filter unit shall be inspected to assure that it is still cycling properly and all circuit breakers are set. Then:

(a) With the power applied as specified in 4.4.6.2(a), disconnect the solenoid/timer apparatus and connect a manometer and adjustable vacuum source (free of pressure pulsations) to the hose fitting on the compartment control module. With zero differential pressure applied to the fitting, the airflow valve shall drive fully open, the horn will sound and the MASK indicator light will flash.
(b) Slowly increase the differential pressure. The horn shall silence and the MASK indicator shall go out. The airflow valve shall begin to drive closed as specified.
(c) Continue to increase the differential pressure to approximately 2.2 iwg until the airflow valve is fully closed. Slowly decrease the differential pressure. The airflow valve shall begin to drive open as specified.
(d) Continue to decrease the differential pressure. The airflow valve shall continue

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to drive open, the horn shall sound and the MASK indicator shall light as specified.

(e) Measure the rotational speed of the fan assembly with a stroboscope. The rotational speed shall be as specified.

5. PACKAGING

5.1 Inter-plant shipment (see 6.4). Military packaging as specified in 5.2 is not applicable to interplant shipments. The filter units shall be packaged to provide adequate protection from physical damage, and marked to identify the filter units and provide any other required data during shipment from the source to the first receiving activity.

5.2 Military packaging. For all shipments, other than interplant shipments (see 6.4), packaging (preservation, unit packing, packing, and marking) shall be as specified in SPI P5-19-6699.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M59 gas-particulate filter unit covered by this specification is designed to maintain a positive internal pressure in vans, vehicles and shelters using a filtered airflow to prevent the infiltration of hazardous levels of chemical and biological agents.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification

(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)

(c) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(d) Level of packing required

6.3 Airflow valve. The terms open and closed, used in this specification to describe the position of the airflow valve, refer to the port marked “TO SHELTER”. The valve door

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blocks the “TO SHELTER” port in the closed position. In the open position, the door blocks the “TO PROT ENT” port (i.e. the. “TO SHELTER” port is not blocked).

6.4 Interplant shipment. Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or between contractors and subcontractors, or from a vendor or contractor to a military arsenal, plant, or other activity for evaluation immediate use. or further processing as specified in the applicable contract.

6.5 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Airflow valve
Compartment control module
Filter unit

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A008

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d. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51539A(EA)	FILTER UNIT, GAS-PARTICULATE: 400 CFM, 208 V, 400 Hz, M59

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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